

Practice Quiz 04A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 04

Set identities by logical laws; unions of families and power sets.

Name a definition or law at every step of part (a).

1. (a) Prove $(A \cup B) \setminus B = A \setminus B$ by a chain of equivalences beginning with membership in the left side.

Solution For an arbitrary object x , the definitions give

$$x \in (A \cup B) \setminus B \iff (x \in A \vee x \in B) \wedge x \notin B.$$

The distributive law gives

$$(x \in A \vee x \in B) \wedge x \notin B \iff (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin B).$$

By the contradiction law, the last statement is equivalent to $x \in A \wedge x \notin B$, which is equivalent by definition to $x \in A \setminus B$. The sets have exactly the same elements, so they are equal. ■

- (b) Let $\mathcal{F} = \{\{0\}, \{0, 1\}\}$. Compute $\bigcup \mathcal{F}$ and then $\mathcal{P}(\bigcup \mathcal{F})$.

Solution $\bigcup \mathcal{F} = \{0, 1\}$, so

$$\mathcal{P}(\bigcup \mathcal{F}) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}. \quad \blacksquare$$